FINAL PROJECT DOCUMENT

UCab – MERN Stack Cab Booking System

Submitted By

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Abstract

UCab is a web-based cab booking application developed using the MERN Stack (MongoDB, Express.js, React.js, and Node.js). The project provides a simple and user-friendly platform where users can register, log in securely, book cabs, view ride details, and access their booking history. The application demonstrates the integration of modern web technologies to build a responsive and dynamic transportation service platform.

The system emphasizes authentication, efficient data management, and an intuitive user interface while following a modular client-server architecture. UCab offers a practical solution for understanding full-stack web application development and serves as a strong foundation for future enhancements such as real-time ride tracking, online payments, driver management, and AI-based route optimization.

Introduction

Transportation plays an essential role in everyday life, and online cab booking systems have simplified the process of hiring vehicles. Traditional booking methods are time-consuming and lack transparency. UCab addresses these challenges by providing an online platform where users can conveniently book rides through a web application.

The project is built using the MERN Stack, which enables seamless communication between the frontend, backend, and database. The application offers a secure authentication system, efficient ride management, and an interactive interface suitable for modern users.

Objectives

The primary objectives of the UCab project are:

- To develop a full-stack cab booking application using the MERN Stack.
- To provide secure user authentication using JWT.
- To enable users to register and log in securely.
- To allow users to book cabs with calculated fares.
- To maintain booking history for each user.
- To design a responsive and user-friendly interface.
- To demonstrate practical implementation of client-server architecture.

Technology Stack

Frontend

- React.js
- HTML5
- CSS3
- JavaScript

Backend

- Node.js
- Express.js

Database

- MongoDB
- Mongoose

Authentication

- JSON Web Token (JWT)

Development Tools

- Visual Studio Code
- MongoDB Compass
- GitHub
- ThunderClient

System Architecture

The UCab application follows a three-tier architecture:

Presentation Layer

- React.js user interface

Application Layer

- Node.js and Express.js server
- Authentication and API handling

Database Layer

- MongoDB stores user accounts and booking information

The frontend communicates with backend APIs, which interact with the MongoDB database to process user requests and manage ride information.

Modules

User Registration

Allows new users to create an account by providing required information.

Features

- User signup
- Input validation
- Secure password storage

User Login

Allows registered users to authenticate securely.

Features

- JWT Authentication

- Session management
- Secure login

Cab Booking

Allows authenticated users to book rides.

Features

- Enter pickup location
- Enter destination
- Fare calculation
- Ride confirmation

Booking History

Displays previously booked rides.

Features

- View booking details
- Ride history management

Working Procedure

1. User opens the UCab application.
2. User registers if new.
3. Existing users log in securely.
4. User enters pickup and destination locations.
5. The system calculates the ride fare.
6. Booking details are saved in MongoDB.
7. User can view booking history.

Features

- Secure user authentication
- User registration

- Login system
- Cab booking
- Automatic fare calculation
- Booking history
- Responsive interface
- MERN Stack architecture
- REST API integration
- MongoDB database storage

Advantages

- Simple and user-friendly interface
- Secure authentication using JWT
- Fast data retrieval with MongoDB
- Responsive design
- Easy to maintain and extend
- Demonstrates complete MERN Stack development
- Modular project structure

Limitations

- No real-time GPS tracking
- No online payment integration
- Driver management module is not included
- No ride cancellation functionality
- Basic fare calculation
- No notification system

Future Scope

The project can be enhanced with several advanced features, including:

- Real-time GPS tracking using Google Maps API.

- Online payment gateway integration.
- Driver registration and management.
- Ride scheduling and advance booking.
- AI-based fare prediction.
- AI-powered route optimization.
- Emergency SOS feature.
- Push notifications.
- Ratings and reviews.
- Admin dashboard with analytics.
- Mobile application development using React Native.
- Multi-language support.

Testing

The application was tested for various functionalities including:

- User registration
- Login authentication
- Ride booking
- Fare calculation
- Booking history retrieval
- API connectivity
- Database operations

All major functionalities worked as expected under normal operating conditions.

Results

The UCab project successfully demonstrates a complete MERN Stack web application capable of handling user authentication, cab booking, and booking history management. The integration of React.js, Node.js, Express.js, and MongoDB ensures smooth communication between the frontend and backend while maintaining data consistency and application responsiveness.

Conclusion

The UCab project successfully achieved its objective of developing a full-stack cab booking system using the MERN Stack. The application provides a secure, responsive, and efficient platform for users to register, log in, and book cab rides with ease. The implementation of JWT authentication, RESTful APIs, and MongoDB database management demonstrates the practical application of modern web development technologies.

Although the project represents a basic version of a real-world cab booking platform, it establishes a strong foundation for future improvements such as real-time tracking, payment gateway integration, AI-based route optimization, and enhanced user experience. Overall, UCab serves as an effective learning project that showcases the concepts of full-stack web development and modern software engineering practices.